

## Chapter 3.4-3.5 - Fouriers Method I

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### 3B.29

A test of water taken in a thin long cylindrical container of length  $L$ . At the start, the container only has water. At  $t = 0$ , one end of the cylinder is opened, where the substance concentration is  $q_0$ . Solve the diffusion problem.

#### Solution:

Diffusion equation

$$\begin{aligned}\frac{\partial u}{\partial t} - D \frac{\partial^2 u}{\partial x^2} &= 0, \quad 0 < x < L, t > 0 \\ u(0, t) &= q_0, \quad t > 0 \\ \frac{\partial u(L, t)}{\partial x} &= 0, \quad t > 0 \\ u(x, 0) &= 0, \quad 0 < x < L\end{aligned}$$

We start by making the equations homogenous. Introduce

$$v(x, t) = u(x, t) - q_0$$

This gives the model in terms of  $v(x)$ :

$$\begin{aligned}\frac{\partial v}{\partial t} - D \frac{\partial^2 v}{\partial x^2} &= 0, \quad 0 < x < L, t > 0 \\ v(0, t) &= 0, \quad t > 0 \\ \frac{\partial v(L, t)}{\partial x} &= 0, \quad t > 0 \\ v(x, 0) &= -q_0, \quad 0 < x < L\end{aligned}$$

Expressed in terms of a linear operator, this becomes

$$\begin{aligned}\frac{\partial v}{\partial t} + D\mathcal{A}v &= 0, \quad D\mathcal{A} = \{v \in \mathcal{C}^2[0, L] \mid v(0) = 0, v'(L) = 0\} \\ \mathcal{A} &= -\frac{\partial^2}{\partial x^2}\end{aligned}$$

We now want to find the eigenvalues and eigenfunctions of the operator  $\mathcal{A}$

$$\begin{aligned}\mathcal{A}\varphi &= \lambda u \rightarrow \frac{\partial^2 \varphi}{\partial x^2} + \lambda \varphi = 0 \rightarrow \varphi(x) = c_k e^{i\sqrt{\lambda_k}x} = \\ &= A_k \cos(\sqrt{\lambda_k}x) + B_k \sin(\sqrt{\lambda_k}x) \\ &\quad \downarrow \\ \varphi(0) &= A_k \cos(\sqrt{\lambda_k}x) = 0 \rightarrow A_k = 0 \\ \varphi'(L) &= \sqrt{\lambda_k} B_k \cos(\sqrt{\lambda_k}L) = 0 \rightarrow \sqrt{\lambda_k}L = \frac{\pi}{2} + k\pi\end{aligned}$$

This gives

$$\begin{aligned}\lambda_k &= \left(\frac{\pi}{L} \left[\frac{1}{2} + k\right]\right)^2 \\ \varphi_k(x) &= \sin\left(\frac{\pi}{L} \left[\frac{1}{2} + k\right]x\right), \quad k = 0, 1, 2, \dots\end{aligned}$$

and we have found the eigenvalues and eigenfunctions of  $\mathcal{A}$

We now have

$$v(x, t) = \sum_{k=0}^{\infty} v_k(t) \varphi_k(x)$$

Putting this into the differential equation for  $v(x, t)$  gives

$$\begin{aligned}\sum_{k=0}^{\infty} (v'_k(t) + D\lambda_k v_k(t)) \varphi_k(x) &= 0 \\ &\quad \downarrow \\ v'_k(t) + D\lambda_k v_k(t) &= 0 \\ &\quad \downarrow \\ v_k(t) &= c_k e^{-D\lambda_k t}\end{aligned}$$

Now we need to find the coefficients  $c_k$ , which is done by calculating  $v(x, 0)$ :

$$v(x, 0) = \sum_{k=0}^{\infty} c_k e^{-D\lambda_k \cdot 0} \varphi_k(x) = \sum_{k=0}^{\infty} c_k \varphi_k(x) = -q_0$$

The coefficients are now found by using the projection formula

$$\begin{aligned}c_k &= \frac{(\varphi_k | -q_0)}{(\varphi_k | \varphi_k)} = \frac{-q_0 \int_0^L \sin(\sqrt{\lambda_k}x) dx}{\int_0^L \sin^2(\sqrt{\lambda_k}x) dx} = \\ &= q_0 \frac{\left[\frac{\cos(\sqrt{\lambda_k}x)}{\sqrt{\lambda_k}}\right]_0^L}{\left[\frac{1}{2}x - \frac{\sin(2\sqrt{\lambda_k}x)}{4\sqrt{\lambda_k}}\right]_0^L} = q_0 \frac{\frac{\cos(\sqrt{\lambda_k}L)}{\sqrt{\lambda_k}} - \frac{1}{\sqrt{\lambda_k}}}{\frac{L}{2}} = \boxed{-\frac{2q_0}{L\sqrt{\lambda_k}} = v_k(0)}\end{aligned}$$

This gives

$$\begin{aligned}
 v_k(t) &= -\frac{2q_0}{L\sqrt{\lambda_k}} e^{-D\lambda_k t} \\
 &\downarrow \\
 v(x, t) &= -\frac{2q_0}{L} \sum_{k=0}^{\infty} \frac{1}{\sqrt{\lambda_k}} e^{-D\lambda_k t} \sin(\sqrt{\lambda_k} x) \\
 &\downarrow \\
 u(x, t) &= v(x, t) + q_0 = q_0 - \frac{2q_0}{L} \sum_{k=0}^{\infty} \frac{1}{\sqrt{\lambda_k}} e^{-D\lambda_k t} \sin(\sqrt{\lambda_k} x)
 \end{aligned}$$

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### 3B.30

**Solution:**

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### 3B.32a

**Solution:**

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### 3B.35

Determine the modes and eigenfrequencies for sound waves in an exponential horn described by letting

$$y = ke^{ax}, \quad 0 \leq x \leq L$$

rotate about the x-axis. The horn is open in both ends. The wave equation for a variable area is given by

$$\frac{\partial^2 u}{\partial t^2} - v^2 \frac{1}{A(x)} \frac{\partial}{\partial x} \left( A(x) \frac{\partial u}{\partial x} \right) = 0$$

**Solution:**

The area of the horn is

$$A(x) = \pi y^2 = \pi k^2 e^{2ax}$$

The differential equation can be described by a linear Sturm-Liouville operator

$$\mathcal{A} = -\frac{1}{A(x)} \frac{\partial}{\partial x} \left( A(x) \frac{\partial \varphi}{\partial x} \right)$$

$$D_{\mathcal{A}} = \{ \varphi \in \mathcal{C}^2[0, L] \mid \varphi(0) = \varphi(L) = 0 \}$$

Find eigenvalues and eigenfunctions to the operator

$$\begin{aligned} \mathcal{A}\varphi = \lambda\varphi &\rightarrow -\frac{1}{A(x)} \frac{\partial}{\partial x} \left( A(x) \frac{\partial \varphi}{\partial x} \right) = \lambda\varphi(x) \\ &\downarrow \\ -\frac{1}{A(x)} [A'(x)\varphi'(x) + A(x)\varphi''(x)] &= \lambda\varphi(x) \\ &\downarrow \\ -2a\varphi'(x) - \varphi''(x) = \lambda\varphi(x) &\rightarrow \varphi''(x) + 2a\varphi'(x) + \lambda\varphi(x) \end{aligned}$$

This gives the characteristic equation

$$r^2 + 2ar + \lambda = 0 \rightarrow r = -a \pm \sqrt{a^2 - \lambda}$$

This gives three cases for the square root:

$$\begin{cases} (1): a^2 - \lambda > 0 \\ (2): a^2 - \lambda = 0 \\ (3): a^2 - \lambda < 0 \end{cases}$$

For the first case:

$$\begin{aligned} (1): r &= -a \pm \sqrt{a^2 - \lambda} \\ &\downarrow \\ \varphi(x) &= c_1 e^{(-a+\sqrt{a^2-\lambda})x} + c_2 e^{(-a-\sqrt{a^2-\lambda})x} \\ \varphi(0) &= c_1 + c_2 = 0 \rightarrow c_1 = -c_2 \\ \varphi(L) &= c_1 e^{(-a+\sqrt{a^2-\lambda})L} - c_1 e^{(-a-\sqrt{a^2-\lambda})L} \rightarrow e^{-\sqrt{a^2-\lambda}} - e^{\sqrt{a^2-\lambda}} = 0 \end{aligned}$$

This is not possible, hence  $c_1 = c_2 = 0$  and this corresponds to no eigenvalues or eigenfunctions.

Case 2:

$$(2): r = -a$$

$$\downarrow$$

$$\varphi(x) = (c_1 + c_2 x)e^{-ax}$$

$$\varphi(0) = c_1 = 0 \rightarrow c_1 = 0$$

$$\varphi(L) = c_2 L e^{-aL} = 0 \rightarrow c_2 = 0$$

This also gives no non-trivial solution as both  $c_1$  and  $c_2$  are zero.

Case 3:

$$(3): r = -a \pm i\sqrt{\lambda_k - a^2} = -a \pm i\omega_k, \text{ with } \omega_k = \sqrt{\lambda_k - a^2}$$

$$\downarrow$$

$$\varphi(x) = e^{-ax} (A \cos(\omega_k x) + B \sin(\omega_k x))$$

$$\varphi(0) = A \cos(\omega_k 0) \rightarrow A = 0$$

$$\varphi(L) = e^{-aL} \sin(\omega_k L) = 0 \rightarrow \omega_k L = \sqrt{\lambda_k - a^2} = k\pi$$

This gives the eigenvalues and eigenfunctions

$$\lambda_k = \frac{k^2 \pi^2}{L^2} + a^2$$

$$\varphi_k(x) = e^{-ax} \sin\left(\frac{k\pi x}{L}\right), \quad k = 1, 2, 3, \dots$$

$k$  cannot be zero because  $\lambda$  has to be larger than  $a^2$ , and  $k = 0$  gives  $\lambda_0 = a^2$ .

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### 3B.39

**Solution:**

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### 3B.41

**Solution:**

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**3B.42**

**Solution:**

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**3B.43+60**

**Solution:**

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**3B.45a**

**Solution:**

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**3B.46**

**Solution:**

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**3B.49a**

**Solution:**

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**3B.51**

**Solution:**

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**3B.55**

**Solution:**

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**3B.62**

**Solution:**

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**3B.66**

**Solution:**

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**3B.69**

**Solution:**

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